Operating Systems Placement Notes: Program, Process & PCB

1. Program vs Process

Program: Passive instructions stored on disk.
Process: Active executing instance of a program in RAM.
Key Difference: Program is static; Process is dynamic and has resources.

2. Process Resources

CPU time, RAM, Code Segment, Data Segment, BSS, Heap, Stack, Open Files, Network Sockets, I/O Devices.

3. Process Memory Layout

Code/Text: Instructions.
Data: Initialized global/static variables.
BSS: Uninitialized global/static variables.
Heap: Dynamic memory.
Stack: Local variables & function calls.

4. Process States

New → Ready → Running → Waiting → Ready → Running → Terminated.

5. PCB

Kernel data structure storing process information for management and context switching.

6. PCB Fields

PID: Unique process ID.
Process State: Current state.
Program Counter: Address of next instruction.
CPU Registers: Saved register values.
Scheduling Info: Priority, queue, time slice.
Memory Info: Page table/base-limit/memory map.
Accounting Info: CPU time, user ID, usage stats.
I/O Status: Open files & devices.
Parent/Child Info: Process relationships.

7. PCB vs Resources

PCB stores information/references; resources are the actual CPU, memory, files, etc.

8. Context Switching

Save current process context into PCB, load next PCB, resume execution.

Frequently Asked Interview Questions & Answers

Q: What is a process?

A: An executing instance of a program.

Q: Program vs Process?

A: Program is passive on disk; process is active in RAM.

Q: What is PCB?

A: A kernel data structure storing information about a process.

Q: Who creates PCB?

A: The Operating System.

Q: Why is PCB needed?

A: To manage processes and perform context switching.

Q: What is PID?

A: A unique identifier for each process.

Q: What is Program Counter?

A: It stores the address of the next instruction to execute.

Q: Ready vs Waiting?

A: Ready waits for CPU; Waiting waits for an event/I/O.

Q: What is stored in Data Segment?

A: Initialized global and static variables.

Q: Difference between Heap and Stack?

A: Heap is dynamic memory; Stack stores local variables and function calls.

Q: Does PCB store actual resources?

A: No, it stores information/references to resources.

Q: What happens when a process terminates?

A: Resources are released and PCB is removed.